TEXORA · LECTURE SHEET

Sliver, Roving and Yarn Irregularity

No spinning process makes a perfectly even strand. Irregularity is the variation of mass along the strand — measured as U% and CV%, displayed as diagrams and spectrograms, and itemised as thin places, thick places and neps. This sheet is the diagnostic heart of yarn quality control.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 13 · LEVEL 3 TERM 1

1. What irregularity is

Irregularity (unevenness) is the variation of a strand's mass per unit length (equivalently, of its cross-section) about its mean, measured over a defined length scale. Even a perfect drafting process cannot eliminate it: fibres arrive randomly, and a strand assembled from a finite number of fibres obeys Poisson-type statistics — the **limit irregularity**. For a strand of mean $\bar{n}$ fibres per cross-section, the theoretical minimum is:

$$\text{Limit CV \%} \approx 100 / \sqrt{\bar{n}}$$

Real strands always exceed the limit (index of irregularity I = actual CV / limit CV, typically 1.2–2+ in spun yarns): drafting adds its own errors on top of the random arrangement. Everything in process control is an attempt to approach the limit, and every fault pushes away from it in a recognisable pattern.

2. How irregularity is measured — the capacitance evenness tester

The standard instrument (the Uster evenness tester family) draws the strand at constant speed (yarn commonly at 400 m/min for routine tests) through a **capacitance measuring slot**: the strand's mass between the plates changes the capacitance in proportion to its mass per unit length. The signal is processed into:

- **The mass diagram** — a trace of mass variation against length (Figure 1): the eye reads medium- and long-term variation, steps and periodic waves directly from it.
- **U% and CV%** — summary statistics (see §3).
- **The spectrogram** — the variation decomposed by wavelength (see §4).
- **Imperfection counts** — thin places, thick places and neps per km at chosen thresholds (see §5).

- **Variance–length curves** — CV measured between different cut lengths, separating short-term from long-term variation (used mainly on sliver and roving).

Optical (diameter-based) testers complement capacitance: diameter variation and mass variation differ where the strand's packing density changes, and comparing the two exposes faults such as hairiness-driven diameter noise (Topic 14).

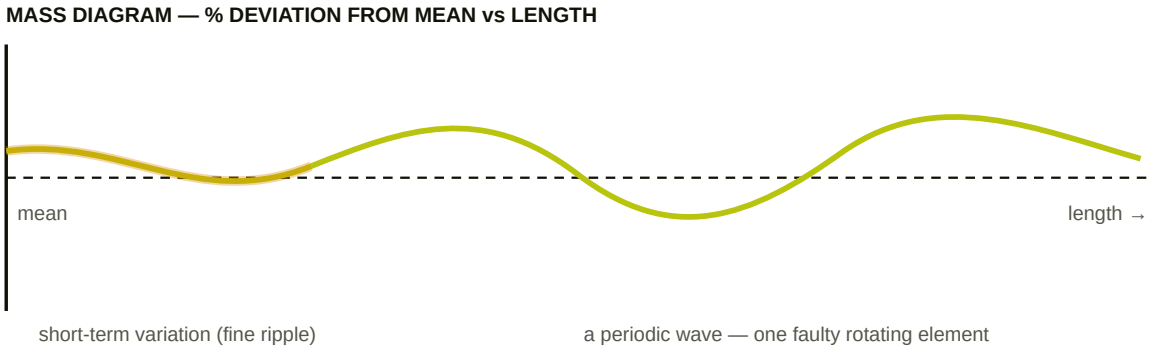


Figure 1 — Evenness-tester mass diagram (schematic). Random ripple is normal; a regular wave repeating at one wavelength is a machine fault announcing its own wavelength — the spectrogram (§4) measures that wavelength exactly.

3. U% and CV% — the two statistics

U% = mean absolute deviation of mass from the mean, as % of the mean (mean deviation)

CV% = standard deviation of mass, as % of the mean

For a normal (Gaussian) distribution of mass values the two are linked by a constant:

$$CV\% \approx 1.25 \times U\%$$

WORKED EXAMPLE — U%, CV% AND THE LIMIT IRREGULARITY

(a) A yarn tests at U% = 12.0 %. Expected CV% ≈ 1.25 × 12.0 = **15.0 %**. If the instrument instead reports CV 17.5 % at that U, the mass distribution is not normal — it has heavy tails (faults) — and the imperfection counts will show it.

(b) The yarn is 19.7 tex cotton of mean fibre fineness 170 mtex. Fibres per cross-section $\bar{n} \approx 19.7 / 0.170 = 116$. Limit CV ≈ 100/√116 = **9.3 %**. The measured CVm of 15.0 % gives an index of irregularity I = 15.0/9.3 = **1.61** — the process is adding substantial irregularity beyond the random limit; there is something to find and fix.

Judge values against percentile statistics for the yarn type and count (the instrument maker's world statistics are the usual benchmark): a carded Ne 30 ring yarn around U% 12–14 is ordinary commercial

work; combed fine yarns run several points lower. Numbers without the count and route are uninterpretable.

4. The spectrogram — finding the faulty element

The **spectrogram** plots the amplitude of variation against its **wavelength** (log scale, from centimetres to hundreds of metres). Reading it:

- **A broad hump** at short wavelengths (drafting wave region) — fibres floating uncontrolled in a drafting zone: roller gauge too wide for the fibre length (Topic 3), insufficient roller pressure, wrong draft distribution.
- **A sharp peak ("chimney")** at one wavelength — a periodic fault from a rotating element. The element's circumference (or circumference × the draft between it and delivery) equals the wavelength.
- **Rising long-wavelength end** — slow drift: count drift, piecing faults, doff-to-doff variation.

WORKED EXAMPLE — WAVELENGTH NAMES THE ROLLER

A roving spectrogram shows a sharp peak at wavelength 12.6 cm at the delivery of the roving frame. The front roller of the preceding draw frame has diameter 2.8 cm → circumference $\pi \times 2.8 = 8.8$ cm — not a match. The draw-frame *middle* roller's fault is magnified by the draft between it and delivery: if that draft is 1.43, its signature appears at $8.8 \times 1.43 =$ **12.6 cm** ✓. An eccentric or damaged middle roller at the draw frame is the suspect — the spectrogram has converted a yarn fault into a work order.

5. Imperfections — thin places, thick places, neps

CLASS	STANDARD THRESHOLDS (CAPACITANCE, MASS)	TYPICAL ORIGIN
Thin places	−40 % and −50 % of mean mass (reported per km)	Drafting faults, fibre shortage, fly loss; the −50 % class predicts spinning breaks
Thick places	+35 % and +50 % of mean mass (per km)	Undrafted tufts, piecings, foreign fibre, drafting surges
Neps (yarn)	+200 % of mean mass over the tester's short reference length (per km)	Fibre neps surviving from the material (Topic 6) — a yarn nep count is largely an AFIS nep count that escaped

Counts are always quoted **per kilometre** and at the stated sensitivity: "thick places 180" with no threshold or length is not a result.

6. Sliver and roving control

- **Sliver:** short test lengths on the evenness tester, or cut-and-weigh of 1 m / 5 m lengths (Topic 10) for long-term weight control; autolevellers at the card/draw frame correct medium-term variation, and their working is verified by these tests (A% — the percentage correction the autoleveller is applying — diagnoses feed variation).
- **Roving:** spectrograms catch roller faults before they multiply through the spinning draft; roving CV between positions exposes individual faulty heads.
- **The doubling rule:** doubling d strands reduces the random component of irregularity by √d — which is why draw frames double 6–8 slivers, and why no single-passage process can match it for long-term evenness.

Glossary

Irregularity — variation of mass per unit length along a strand. **U%** — mean-deviation unevenness. **CV%** — standard-deviation unevenness (≈ 1.25 × U% for normal distributions). **Limit irregularity** — the Poisson minimum, 100/√n̄ %. **Index of irregularity** — actual CV ÷ limit CV. **Spectrogram** — variation amplitude plotted against wavelength. **Drafting wave** — short-wavelength irregularity from floating fibres in drafting. **Imperfections** — thin places, thick places and neps counted per km at set thresholds. **Autoleveller** — a device that varies draft continuously to correct sliver mass variation.

Key points

• Irregularity is mass variation; its floor is set by fibre statistics (100/√n̄), its excess by the process.
• CV% ≈ 1.25 × U% when the distribution is normal; disagreement between them signals faults.
• The mass diagram shows it, the spectrogram locates it — wavelength ÷ draft = the guilty element's circumference.
• Imperfections are reported per km at named thresholds (−50 %, +50 %, +200 %).
• Test sliver and roving, not only yarn: faults found early are faults that never reach a loom.

Practice

1. A yarn shows U% = 11.2 %. What CV% do you expect, and what would a reported CV% of 16.5 % tell you?

2. A 14.8 tex yarn is spun from 155 mtex fibres. Estimate the limit CV% and the index of irregularity if the measured CV% is 14.0 %.
3. A yarn spectrogram peaks at 90 cm wavelength. The spinning frame's front roller circumference is 9.0 cm and the total draft after the suspected element is 10. What element signature matches, and what do you inspect?
4. Why does doubling eight slivers at the draw frame not reduce irregularity to one-eighth?

Answer key

1. Expected CV ≈ 1.25 × 11.2 = **14.0 %**. A reported 16.5 % is well above the normal-distribution relation, so the mass distribution has fault tails — expect elevated imperfection counts; investigate periodic / drafting faults rather than accepting the U% as "good".
2. $\bar{n}$ = 14.8/0.155 ≈ 95 fibres. Limit CV ≈ 100/√95 = **10.3 %**. Index I = 14.0/10.3 = **1.36** — reasonably controlled spinning.
3. The wavelength at its source = 90 / 10 = **9.0 cm**. So the guilty element has a circumference of 9.0 cm and sits where a draft of 10 still acts after it — for example the back roller of the spinning-frame drafting system (a fault at the front roller itself would appear at the front roller's own circumference, 9.0 cm, with no multiplication, not at 90 cm). Inspect that back roller and its bearing for eccentricity and damage.
4. Doubling averages independent random variations: the variance of the mean of d strands is 1/d of one strand's, so CV falls by √d (here √8 ≈ 2.83), not by d. Systematic components (a periodic fault present in all strands in phase) are not reduced at all.
